

Early Subclinical Macular Edema in Eyes with Uveal Melanoma: Association with Future Cystoid Macular Edema

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Purpose: To determine the frequency of early subclinical macular edema in eyes with uveal melanoma and its association with future cystoid macular edema (CME).

Design: Retrospective cohort study.

Participants: A total of 306 patients with uveal melanoma; 260 patients had follow-up of 1 or more years after plaque radiotherapy (follow-up cohort).

Methods: Review of medical records and spectral-domain optical coherence tomography (OCT) images.

Main Outcome Measures: Frequency of early subclinical macular edema (increased central macular thickness of $>10\ \mu\text{m}$ without cystoid changes before or at 4 months after plaque radiotherapy); rate of future CME.

Results: At baseline, 164 patients (54%) had subclinical macular edema in the involved eye. On multivariate analysis, factors associated with subclinical macular edema at baseline were increasing tumor diameter ($P = 0.001$), increasing tumor thickness ($P = 0.010$), and subretinal fluid ($P = 0.001$). Of 260 patients in the follow-up cohort, 105 (40%) developed CME during a median follow-up of 31 months (mean, 34; range, 12–70 months). Eyes with subclinical macular edema at baseline (and at 4 months after plaque radiotherapy) had a significantly higher rate of future CME ($n = 66$; 50%) compared with eyes without subclinical macular edema at baseline ($n = 39$; 30%) ($P = 0.005$; hazard ratio, 1.77; 95% confidence interval, 1.19–2.64). On multivariate analysis, the factors associated with future development of CME included female gender ($P = 0.004$), increasing tumor thickness ($P < 0.001$), decreasing tumor distance to foveola ($P = 0.002$), hemorrhage over tumor ($P = 0.017$), and increased CMT of $>10\%$ at baseline in the involved eyes compared with the opposite eyes ($P = 0.012$).

Conclusions: Subclinical macular edema is common in eyes with uveal melanoma before and at 4 months after plaque radiotherapy and is associated with initial larger tumor size. Eyes with early subclinical macular edema are at significantly higher risk for future CME. These findings suggest that tumor-related factors, most likely mediated through proinflammatory cytokines, may play an important role in development of post-radiation CME. *Ophthalmology* 2015;122:1023–1029 © 2015 by the American Academy of Ophthalmology.



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Cystoid macular edema (CME) is a common cause of visual impairment after plaque radiotherapy of uveal melanoma.^{1,2} In a study on 135 patients with uveal melanoma followed for a median of 24 months after I-125 plaque radiation, Horgan et al¹ found evidence of macular edema by time-domain optical coherence tomography (OCT) in 70% of treated eyes. The mean time to onset of macular edema in their study was 12 months, 5 months shorter than the mean time to onset of clinical radiation maculopathy.

The aim of the study by Horgan et al¹ was to identify early changes in macular morphology after plaque radiotherapy of uveal melanoma. However, because of the limited resolution of time-domain OCT, the researchers were not able to accurately and reliably measure early and subtle changes in macular thickness before the development of OCT-evident CME. Compared with time-domain OCT with an axial resolution of $10\ \mu\text{m}$ or more, spectral-domain

OCT has a higher resolution down to 3 to $5\ \mu\text{m}$.³ Since March of 2007, all patients with uveal melanoma managed in our center have been evaluated with spectral-domain OCT before treatment and at each visit after plaque radiotherapy. By taking advantage of the higher resolution of spectral-domain OCT, we performed this study to measure early increase in central macular thickness (CMT) in eyes with uveal melanoma before and after plaque radiotherapy and to determine whether such early changes in CMT are predictive of future development of CME.

Methods

We reviewed the medical records of patients with the diagnosis of uveal melanoma who were treated with I-125 plaque radiotherapy at the Ocular Oncology Service, Wills Eye Hospital between April 2007 and January 2012. Patients with the following conditions